

Phishing and Malicious Email Classification Using LSTM

Deep Learning Applications — ELA-1

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1. Introduction and Problem Statement

Phishing and malicious emails are a major cybersecurity threat because they can deceive users into revealing credentials, financial information, or other sensitive data. Traditional rule-based filters may struggle when attackers change their wording and communication patterns. This project develops a deep learning-based approach that learns patterns from labelled email text and classifies each message as legitimate or phishing/malicious.

Objectives

- Develop an LSTM-based binary email classifier.
- Preprocess and tokenize real-world email text.
- Evaluate the model using accuracy, precision, recall, F1-score and a confusion matrix.

2. Dataset

The project uses the **phishing_email.csv** real-world email dataset containing **38,925 labelled records**. The dataset contains email text and binary labels, where 0 represents legitimate emails and 1 represents phishing/malicious emails. Missing values and duplicate records were handled before model training, followed by text preprocessing and removal of unusable empty records.

Dataset Property	Value
Original samples	38,925
Samples after cleaning	38,525
Final usable samples	38,523
Legitimate (label 0)	20,654
Phishing/Malicious (label 1)	17,871
Train / Test split	80% / 20%

3. Methodology

Data cleaning: Missing values and duplicate records were removed.

Text preprocessing: Text was converted to lowercase, URLs were normalized to a common token, unnecessary characters were removed, and extra whitespace was normalized.

Splitting: An 80:20 stratified train-test split was applied to preserve class proportions.

Tokenization: A Keras tokenizer with a 15,000-word vocabulary was fitted on the training data.

Padding: Sequences were padded or truncated to a fixed length of 150 tokens.

Processing Pipeline

Raw Email → Cleaning → Text Preprocessing → Tokenization → Sequence Padding → Word Embedding → LSTM → Dense Layers → Sigmoid Output

4. Model Architecture and Training

A sequential LSTM neural network was selected because LSTMs can learn sequential dependencies and contextual patterns in email text. The architecture consists of an embedding layer, an LSTM layer, dropout layers for regularization, a ReLU dense layer, and a sigmoid output neuron for binary classification.

Component / Parameter	Configuration
Embedding	15,000-word vocabulary × 64 dimensions
LSTM	64 units
Dropout	0.30 and 0.20
Dense	32 units, ReLU
Output	1 unit, Sigmoid
Optimizer	Adam
Loss	Binary Cross-Entropy
Batch size	64
Maximum epochs	Up to 8 with Early Stopping
Early stopping	Validation loss, patience = 2

5. Training Performance

The model was trained with validation monitoring and early stopping. The training and validation curves show the learning behaviour across epochs. The best validation weights were retained to reduce overfitting and improve generalization on unseen test emails.

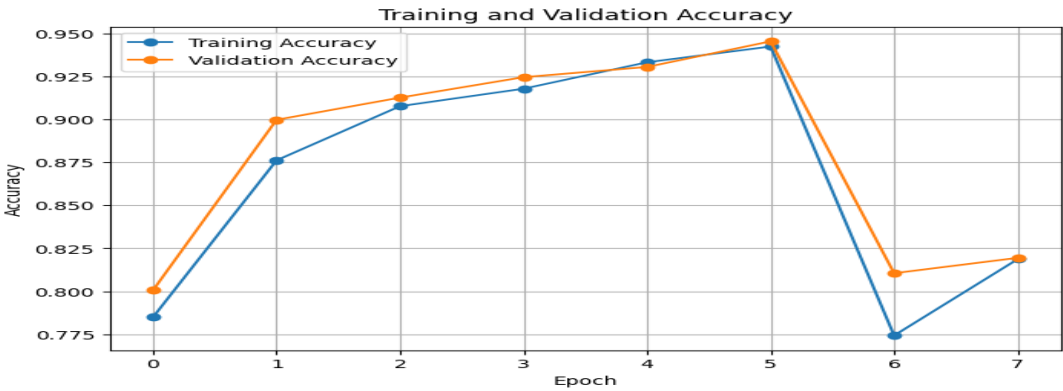


Figure 1. Training and Validation Accuracy

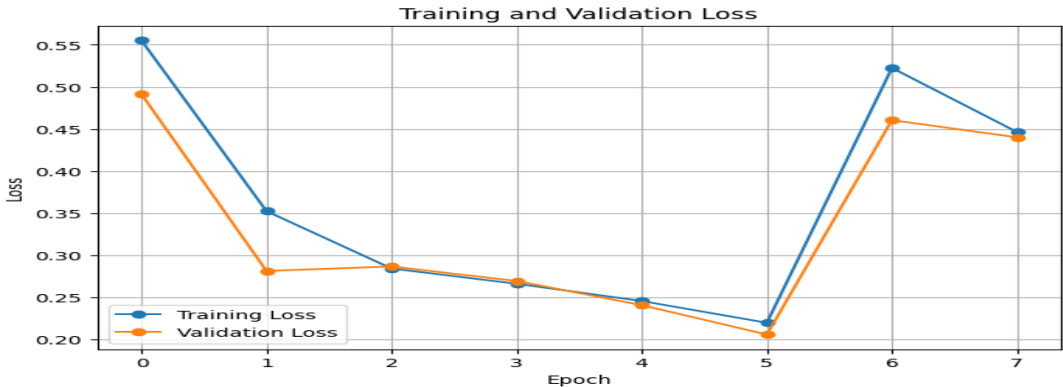


Figure 2. Training and Validation Loss

6. Test Results

The final LSTM model was evaluated on the held-out test set. The model achieved strong overall classification performance, with particularly high recall for phishing emails, indicating that most malicious messages were successfully detected.

Metric	Test Result
Accuracy	94.76%
Precision	84.93%
Recall	91.85%
F1-Score	88.26%
Test Loss	0.1874

7. Confusion Matrix

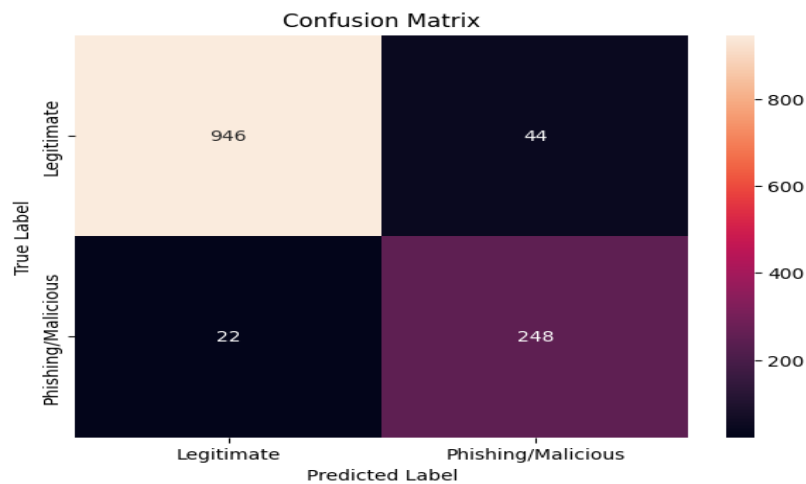


Figure 3. Confusion Matrix on the Held-Out Test Set

The model correctly classified **946 legitimate** emails and **248 phishing/malicious** emails. It produced **44 false positives** and **22 false negatives**. The high recall of 91.85% means that the model successfully detected most phishing emails, while the false-positive count indicates that some legitimate messages may still be flagged incorrectly.

8. Discussion and Limitations

The results show that the LSTM model can effectively learn textual patterns associated with legitimate and phishing emails. However, the dataset may not represent every writing style, organization, language, or newly emerging phishing strategy. The present model also uses only email text and does not consider sender metadata, URL structure, attachments, or other security indicators.

9. Conclusion and Future Scope

This project successfully implemented a deep learning pipeline for phishing and malicious email classification using text preprocessing, tokenization, padding and an LSTM neural network. The final model achieved **94.76% test accuracy** and an **88.26% F1-score**. Future work can use larger and more diverse datasets, Transformer-based models, class-balancing methods, and additional features such as URLs, sender information and attachments to improve real-world email security.